01:XXX:XXX – Homework n

Pranav Tikkawar

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1 Metric Spaces

Definition (Metric Space)**.** A metric on a nonempty set X is a function

$$d : X \times X \longrightarrow [0, \infty)$$

such that, for all $p, q, r \in X$:

1. $d(p, q) = 0$ if and only if $p = q$.
2. $d(p, q) = d(q, p)$.
3. $d(p, r) \leq d(p, q) + d(q, r)$.

The pair (X, d) is called a metric space.

Remark (Metric Subspaces)**.** If Y is a nonempty subset of a metric space (X, d), then

$$d_Y = d|_{Y \times Y}$$

is a metric on Y. The space (Y, d_Y) is called a metric subspace of (X, d), and X is its ambient space.

Definition (Open Interval)**.** For $a, b \in \mathbb{R}$ with $a < b$, define

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}.$$

Definition (Open Ball)**.** Let (X, d) be a metric space. For $p \in X$ and $r > 0$, define

$$B_X(p, r) = \{x \in X : d(x, p) < r\}.$$

When the ambient space is clear, we write $B(p, r)$. Other common notations are $B_r(p)$ and $V_r(p)$.

Definition (Neighborhood in a Metric Space)**.** A set $N \subseteq X$ is a neighborhood of $p \in X$ if there exists $r > 0$ such that

$$B(p, r) \subseteq N.$$

A neighborhood need not itself be open. An open neighborhood is a neighborhood that is open.

Definition (Interior Point)**.** A point $p \in X$ is an interior point of $E \subseteq X$ if there exists $r > 0$ such that

$$B(p, r) \subseteq E.$$

Definition (Interior)**.** The interior of $E \subseteq X$ is the set of all interior points of E:

$$\text{int}_X(E) = \{p \in X : \text{there exists } r > 0 \text{ such that } B(p, r) \subseteq E\}.$$

We also write E° when the ambient space is clear.

Definition (Exterior)**.** The exterior of $E \subseteq X$ is

$$\operatorname{ext}_X(E) = \operatorname{int}_X(X \setminus E).$$

Definition (Open Set in a Metric Space)**.** A set $E \subseteq X$ is open if every point of E is an interior point. Equivalently,

$$E = \operatorname{int}_X(E).$$

Remark (The Metric Topology)**.** The open subsets of a metric space form a topology:

$$\tau_d = \{U \subseteq X : \text{for every } p \in U, \text{ there exists } r > 0 \text{ with } B(p, r) \subseteq U\}.$$

Whenever a metric space is treated as a topological space, we use this topology unless stated otherwise.

2 Topological Spaces and Separation

Definition (Topological Space)**.** Let X be a set. A topology on X is a collection $\tau \subseteq \mathcal{P}(X)$ such that:

1. $\emptyset, X \in \tau$.
2. Any union of members of τ belongs to τ.
3. Any finite intersection of members of τ belongs to τ.

The pair (X, τ) is a topological space. The members of τ are called open sets.

Definition (Neighborhood in a Topological Space)**.** A set $N \subseteq X$ is a neighborhood of $p \in X$ if there exists an open set U such that

$$p \in U \subseteq N.$$

An open neighborhood of p is an open set containing p.

Remark (Neighborhood Conventions)**.** Some texts use the word "neighborhood" to mean only an open neighborhood. These notes allow neighborhoods that are not open and explicitly say "open neighborhood" when openness is required.

Definition (Interior in a Topological Space)**.** For $E \subseteq X$, a point p belongs to $\operatorname{int}_X(E)$ if there exists an open set U such that

$$p \in U \subseteq E.$$

Equivalently, $\operatorname{int}_X(E)$ is the union of all open subsets of E.

Definition (Closed Set)**.** A set $F \subseteq X$ is closed if $X \setminus F$ is open.

Definition (Subspace Topology)**.** If $Y \subseteq X$, the subspace topology on Y is

$$\tau_Y = \{Y \cap U : U \in \tau\}.$$

Thus a subset of Y may be open in Y without being open in X.

Remark (Separation Axioms)**.** The following axioms describe how well a topology separates points and closed sets.

1. T_0: For distinct $p, q \in X$, there exists an open set containing one of them but not the other.

2. T_1: For distinct $p, q \in X$, there exist open sets U, V such that

   $$p \in U, \quad q \notin U, \qquad q \in V, \quad p \notin V.$$

   Equivalently, every singleton is closed.

3. T_2 (Hausdorff): For distinct $p, q \in X$, there exist disjoint open neighborhoods U of p and V of q.

4. T_3: The space is T_1, and whenever F is closed and $p \notin F$, there exist disjoint open sets U, V such that

   $$p \in U, \qquad F \subseteq V.$$

5. T_4: The space is T_1, and whenever F_1, F_2 are disjoint closed sets, there exist disjoint open sets U_1, U_2 such that

   $$F_1 \subseteq U_1, \qquad F_2 \subseteq U_2.$$

With these conventions,

$$T_4 \Longrightarrow T_3 \Longrightarrow T_2 \Longrightarrow T_1 \Longrightarrow T_0.$$

Terminology varies: some authors define "regular" and "normal" without including the T_1 condition.

Definition (Hausdorff Space)**.** A topological space is Hausdorff if it satisfies T_2: distinct points have disjoint open neighborhoods.

Theorem 1 (Metric Spaces Are Hausdorff)**.** *Every metric space is Hausdorff in its metric topology.*

Proof. Let $p \neq q$ and put

$$r = \frac{d(p, q)}{2} > 0.$$

Suppose $x \in B(p, r) \cap B(q, r)$. Then

$$d(p, q) \leq d(p, x) + d(x, q) < r + r = d(p, q),$$

a contradiction. Therefore the two open balls are disjoint. $\square$

3 Closure, Limit Points, and Boundary

Definition (Closure)**.** For $E \subseteq X$, its closure is

$$\overline{E} = \bigcap \{F \subseteq X : F \text{ is closed and } E \subseteq F\}.$$

Equivalently,

$$p \in \overline{E} \quad \Longleftrightarrow \quad \text{every open neighborhood of } p \text{ intersects } E.$$

Definition (Limit Point)**.** A point $p \in X$ is a limit point, or accumulation point, of $E \subseteq X$ if every open neighborhood U of p satisfies

$$U \cap (E \setminus \{p\}) \neq \emptyset.$$

The point p need not belong to E.

Definition (Derived Set)**.** The derived set of E is

$$E' = \{p \in X : p \text{ is a limit point of } E\}.$$

Theorem 2 (Closure and the Derived Set)**.** *In every topological space,*

$$\overline{E} = E \cup E'.$$

Consequently,

$$E \text{ is closed} \quad \Longleftrightarrow \quad E' \subseteq E.$$

Proof. Every point of E belongs to $\overline{E}$. Every point of E' also belongs to $\overline{E}$, because every neighborhood intersects E.

Conversely, suppose $p \in \overline{E} \setminus E$. Every open neighborhood of p intersects E. Since $p \notin E$, such an intersection contains a point of E different from p. Hence $p \in E'$. □

Remark (Closed Sets and Sequence Limits)**.** In every topological space, a closed set contains every limit of every convergent sequence of its points.

The converse holds in metric spaces but can fail in general topological spaces. Thus "contains all its limit points" should not be confused with a characterization using sequences alone in arbitrary spaces.

Definition (Isolated Point)**.** A point $p \in E$ is isolated in E if there exists an open neighborhood U of p such that

$$U \cap E = \{p\}.$$

Equivalently, $p \in E \setminus E'$.

Theorem 3 (Open and Closed Complements)**.** *A set is closed if and only if its complement is open.*

Remark. This is the definition of closedness adopted above. If closedness is instead introduced by $E = \overline{E}$, the same statement becomes an equivalent characterization.

Definition (Boundary)**.** The boundary of $E \subseteq X$ is

$$\partial E = \overline{E} \cap \overline{X \setminus E}.$$

Equivalently, $p \in \partial E$ if every open neighborhood of p intersects both E and $X \setminus E$.

Remark (Boundary Identities)**.** For every $E \subseteq X$,

$$\partial E = \overline{E} \setminus \operatorname{int}_X(E), \qquad \partial E = \partial(X \setminus E).$$

Moreover,

$$X = \operatorname{int}_X(E) \ \cup \ \partial E \ \cup \ \operatorname{ext}_X(E),$$

where the three sets are pairwise disjoint.

Remark (Boundary Points Versus Limit Points)**.** The boundary definition does not exclude p from the intersections.

More precisely:

$$p \in E \quad \Longrightarrow \quad \big(p \in \partial E \Longleftrightarrow p \in (X \setminus E)'\big),$$

and

$$p \notin E \quad \Longrightarrow \quad \big(p \in \partial E \Longleftrightarrow p \in E'\big).$$

A boundary point need not be a limit point of both sets. For example, in the usual topology on $\mathbb{R}$,

$$\partial\{0\} = \{0\}, \qquad \{0\}' = \emptyset.$$

4 Sequences and Completeness

Definition (Convergent Sequence)**.** A sequence (x_n) in a topological space X converges to $x \in X$ if, for every open neighborhood U of x, there exists $N \in \mathbb{N}$ such that

$$n \geq N \quad \Longrightarrow \quad x_n \in U.$$

We write $x_n \to x$.

In a metric space, this is equivalent to: for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \quad \Longrightarrow \quad d(x_n, x) < \varepsilon.$$

Theorem 4 (Uniqueness of Limits in Hausdorff Spaces)**.** *A convergent sequence in a Hausdorff space has a unique limit.*

Proof. Suppose $x_n \to p$ and $x_n \to q$ with $p \neq q$. Choose disjoint open neighborhoods U of p and V of q. Eventually x_n belongs to both U and V, which contradicts $U \cap V = \emptyset$. $\square$

Definition (Cauchy Sequence)**.** A sequence (x_n) in a metric space (X, d) is Cauchy if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$m, n \geq N \quad \Longrightarrow \quad d(x_m, x_n) < \varepsilon.$$

Definition (Complete Metric Space)**.** A metric space (X, d) is complete if every Cauchy sequence in X converges to a point of X.

Remark (Complete Versus Closed)**.** Completeness is not the same definition as closedness. A subset is closed relative to an ambient topology, whereas completeness is a property of a metric space.

Every metric space is closed as a subset of itself, but not every metric space is complete.

A closed subspace of a complete metric space is complete. A complete metric subspace of any metric space is closed.

Definition (Permutation)**.** A permutation of a set A is a bijection

$$\sigma : A \longrightarrow A.$$

Injectivity alone is insufficient when A is infinite. For example, $n \mapsto n + 1$ is injective on $\mathbb{N} = \{1, 2, \ldots\}$ but is not surjective.

5 Compactness

Definition (Open Cover)**.** An open cover of $E \subseteq X$ is a family $\{U_\alpha\}_{\alpha \in A}$ of open subsets of X such that

$$E \subseteq \bigcup_{\alpha \in A} U_\alpha.$$

A finite subcover is a finite selection of these same sets that still covers E.

Definition (Compact Set)**.** A subset $E \subseteq X$ is compact if every open cover of E has a finite subcover.

Equivalently, E is compact as a space with its subspace topology. A space X is compact if it is compact as a subset of itself.

Definition (Sequential Compactness)**.** A subset $E \subseteq X$ is sequentially compact if every sequence in E has a subsequence converging to a point of E.

Definition (Limit Point Compactness)**.** A topological space X is limit point compact if every infinite subset of X has a limit point in X.

Remark (Limit Points in T_1 Spaces)**.** In a T_1 space, p is a limit point of E if and only if every neighborhood of p contains infinitely many points of E.

Without the T_1 assumption, the condition involving infinitely many points is stronger than the limit-point definition given above.

Theorem 5 (Compactness Implies Limit Point Compactness)**.** *Every compact topological space is limit point compact.*

Proof. Suppose X is compact and $S \subseteq X$ is infinite. Assume that S has no limit point in X.

For each $p \in X$, choose an open neighborhood U_p such that

$$U_p \cap (S \setminus \{p\}) = \emptyset.$$

Thus $U_p \cap S$ contains at most one point.

The family $\{U_p : p \in X\}$ covers X, so compactness gives a finite subcover

$$X = U_{p_1} \cup \cdots \cup U_{p_m}.$$

It follows that

$$S = \bigcup_{i=1}^{m} (S \cap U_{p_i})$$

is finite, a contradiction. □

Definition (Total Boundedness)**.** A metric space (X, d) is totally bounded if, for every $r > 0$, there exist finitely many points $p_1, \ldots, p_m \in X$ such that

$$X = \bigcup_{i=1}^{m} B(p_i, r).$$

Lemma 1 (Sequential Compactness Implies Total Boundedness)**.** *Every sequentially compact metric space is totally bounded.*

Proof. Suppose X is not totally bounded. Then there exists $r > 0$ such that no finite collection of open balls of radius r covers X.

Choose $x_1 \in X$. Having chosen $x_1, \ldots, x_n$, choose

$$x_{n+1} \notin \bigcup_{i=1}^{n} B(x_i, r).$$

Then

$$d(x_m, x_n) \geq r \qquad (m \neq n).$$

No subsequence can be Cauchy, and therefore no subsequence can converge. This contradicts sequential compactness. □

Lemma 2 (Lebesgue Number Lemma: Ball Form)**.** *Let (X, d) be a sequentially compact metric space and let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of X. There exists $\delta > 0$ such that, for every $x \in X$, some $\alpha \in A$ satisfies*

$$B(x, \delta) \subseteq U_\alpha.$$

Proof. If $X = \emptyset$, the conclusion is vacuous. Otherwise, suppose no such δ exists. For each $n \in \mathbb{N}$, choose $x_n \in X$ such that

$$B(x_n, 1/n) \not\subseteq U_\alpha \qquad \text{for every } \alpha \in A.$$

By sequential compactness, some subsequence (x_{n_k}) converges to a point $x \in X$. Choose α_0 such that $x \in U_{\alpha_0}$. Since U_{α_0} is open, choose $\varepsilon > 0$ such that

$$B(x, \varepsilon) \subseteq U_{\alpha_0}.$$

For sufficiently large k, both

$$d(x_{n_k}, x) < \varepsilon/2 \qquad \text{and} \qquad 1/n_k < \varepsilon/2$$

hold. If $y \in B(x_{n_k}, 1/n_k)$, then

$$d(y, x) \leq d(y, x_{n_k}) + d(x_{n_k}, x) < 1/n_k + \varepsilon/2 < \varepsilon.$$

Hence

$$B(x_{n_k}, 1/n_k) \subseteq B(x, \varepsilon) \subseteq U_{\alpha_0},$$

a contradiction. □

Theorem 6 (Sequentially Compact Metric Spaces Are Compact)**.** *Every sequentially compact metric space is compact.*

Proof. Let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of X. Choose $\delta > 0$ as in the preceding lemma.

Since X is totally bounded, choose $p_1, \ldots, p_m \in X$ such that

$$X = \bigcup_{i=1}^{m} B(p_i, \delta/2).$$

For each i, choose α_i such that

$$B(p_i, \delta) \subseteq U_{\alpha_i}.$$

Then

$$X = \bigcup_{i=1}^{m} B(p_i, \delta/2) \subseteq \bigcup_{i=1}^{m} U_{\alpha_i}.$$

Thus the original cover has a finite subcover. □

Remark (The Choice of Cover Members Matters)**.** In the preceding proof, merely choosing a cover member containing each center p_i is insufficient. We must choose U_{α_i} to contain the entire ball $B(p_i, \delta)$.

Theorem 7 (Compactness Equivalences in Metric Spaces)**.** *For a metric space (X, d), the following are equivalent:*

1. *X is compact.*

2. *X is sequentially compact.*

3. *X is limit point compact.*

4. *X is complete and totally bounded.*

These equivalences are specific to metric spaces; the three compactness notions need not coincide in arbitrary topological spaces.

Lemma 3 (Lebesgue Number Lemma for a Compact Subset)**.** *Let E be a compact subset of a metric space (X, d), and let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of E by open subsets of X.*

There exists $\delta > 0$ such that, for every $p \in E$, some $\alpha \in A$ satisfies

$$B_X(p, \delta) \subseteq U_\alpha.$$

In particular,

$$B_E(p, \delta) = E \cap B_X(p, \delta) \subseteq E \cap U_\alpha.$$

Proof. If $E = \emptyset$, the conclusion is vacuous. Otherwise, for each $p \in E$, choose $\alpha(p)$ and $r_p > 0$ such that

$$B_X(p, 2r_p) \subseteq U_{\alpha(p)}.$$

The balls $B_X(p, r_p)$ cover E. By compactness, finitely many suffice:

$$E \subseteq \bigcup_{i=1}^{m} B_X(p_i, r_{p_i}).$$

Set

$$\delta = \min_{1 \le i \le m} r_{p_i} > 0.$$

For $q \in E$, choose i with $d(q, p_i) < r_{p_i}$. If $y \in B_X(q, \delta)$, then

$$d(y, p_i) \le d(y, q) + d(q, p_i) < \delta + r_{p_i} \le 2r_{p_i}.$$

Therefore

$$B_X(q, \delta) \subseteq B_X(p_i, 2r_{p_i}) \subseteq U_{\alpha(p_i)}.$$

□

Remark (Subspace Covers and Lebesgue Numbers)**.** If the cover consists of sets open only in E, the appropriate conclusion is

$$B_E(p, \delta) \subseteq U_\alpha,$$

not necessarily

$$B_X(p, \delta) \subseteq U_\alpha.$$

A common alternative definition of a Lebesgue number δ requires that every subset of diameter less than δ be contained in one cover member. The ball formulation implies this diameter formulation. Conversely, the diameter formulation gives the ball formulation after shrinking the radius, for example to $\delta/3$.

6 Continuity and Uniform Continuity

Definition (Continuity at a Point)**.** Let (X, d) and (Y, ρ) be metric spaces, and let $f : X \to Y$. The function f is continuous at $p \in X$ if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$d(p, q) < \delta \quad \Longrightarrow \quad \rho(f(p), f(q)) < \varepsilon \qquad (q \in X).$$

Equivalently,

$$f(B_X(p, \delta)) \subseteq B_Y(f(p), \varepsilon).$$

Definition (Continuity)**.** A function $f : X \to Y$ is continuous if it is continuous at every point of X.

For topological spaces, continuity means that $f^{-1}(V)$ is open in X for every open set $V \subseteq Y$. For metric spaces, this agrees with the ε–δ definition.

Remark (Continuity on a Subset)**.** For $A \subseteq X$, distinguish:

1. f is continuous at every $p \in A$ as a function defined on X.
2. The restriction $f|_A : A \to Y$ is continuous with respect to the subspace metric.

The first implies the second, but the converse need not hold. In these notes, "continuous on A" means that $f|_A$ is continuous unless stated otherwise.

Definition (Uniform Continuity)**.** A function $f : X \to Y$ is uniformly continuous on $A \subseteq X$ if

$$\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall p, q \in A : \quad d(p, q) < \delta \Longrightarrow \rho(f(p), f(q)) < \varepsilon.$$

The same δ must work for every pair $p, q \in A$.

Remark (Continuity Versus Uniform Continuity)**.** For continuity at p, the choice of δ may depend on both ε and p.

For uniform continuity, δ may depend on ε, but not on the points p, q.

Remark (Negation of Uniform Continuity)**.** A function f is not uniformly continuous on A if

$$\exists \varepsilon_0 > 0 \; \forall \delta > 0 \; \exists p, q \in A : \quad d(p, q) < \delta \quad \text{and} \quad \rho(f(p), f(q)) \geq \varepsilon_0.$$

Equivalently, there exist sequences $(p_n), (q_n)$ in A such that

$$d(p_n, q_n) \to 0, \qquad \rho(f(p_n), f(q_n)) \geq \varepsilon_0 \quad \text{for every } n.$$

Theorem 8 (Sequential Criterion for Continuity)**.** *Let X, Y be metric spaces and $f : X \to Y$. Then f is continuous if and only if*

$$x_n \to x \quad \Longrightarrow \quad f(x_n) \to f(x)$$

for every sequence (x_n) in X and every $x \in X$.

Proof. Suppose f is continuous and $x_n \to x$. For $\varepsilon > 0$, continuity at x gives $\delta > 0$ such that

$$d(y, x) < \delta \Longrightarrow \rho(f(y), f(x)) < \varepsilon.$$

Eventually $d(x_n, x) < \delta$, so eventually $\rho(f(x_n), f(x)) < \varepsilon$.

Conversely, suppose the sequential condition holds but f is not continuous at some $x \in X$. There exists $\varepsilon_0 > 0$ such that, for every $n \in \mathbb{N}$, we can choose x_n with

$$d(x_n, x) < 1/n, \qquad \rho(f(x_n), f(x)) \geq \varepsilon_0.$$

Then $x_n \to x$, but $f(x_n)$ does not converge to $f(x)$, a contradiction. □

Remark (General Topological Spaces)**.** Continuous maps preserve sequence convergence in every topological space. The converse sequential criterion need not hold for arbitrary topological domains.

Theorem 9 (Continuous Images of Compact Spaces)**.** *If X is a compact topological space and $f : X \to Y$ is continuous, then $f(X)$ is compact.*

Proof. Let $\{V_\alpha\}_{\alpha \in A}$ be a cover of $f(X)$ by open subsets of Y. Then $\{f^{-1}(V_\alpha)\}_{\alpha \in A}$ is an open cover of X. Choose a finite subcover:

$$X = \bigcup_{i=1}^{m} f^{-1}(V_{\alpha_i}).$$

It follows that

$$f(X) \subseteq \bigcup_{i=1}^{m} V_{\alpha_i}.$$

□

Theorem 10 (Heine–Cantor Theorem)**.** *If (X, d) is a compact metric space, (Y, ρ) is a metric space, and $f : X \to Y$ is continuous, then f is uniformly continuous.*

Proof. Suppose f is not uniformly continuous. There exist $\varepsilon_0 > 0$ and sequences $(p_n), (q_n)$ in X such that

$$d(p_n, q_n) < 1/n, \qquad \rho(f(p_n), f(q_n)) \geq \varepsilon_0.$$

By sequential compactness, some subsequence p_{n_k} converges to $p \in X$.

The triangle inequality gives

$$d(q_{n_k}, p) \leq d(q_{n_k}, p_{n_k}) + d(p_{n_k}, p) \to 0,$$

so $q_{n_k} \to p$ as well. By continuity,

$$f(p_{n_k}) \to f(p), \qquad f(q_{n_k}) \to f(p).$$

Consequently,

$$\rho(f(p_{n_k}), f(q_{n_k})) \leq \rho(f(p_{n_k}), f(p)) + \rho(f(p), f(q_{n_k})) \to 0,$$

contradicting the lower bound ε_0. □

7 Sequences of Functions

Remark (Setup)**.** Let (X, d) and (Y, ρ) be metric spaces, and let

$$f_n : X \to Y, \qquad n \in \mathbb{N},$$

be a sequence of functions. Let $f : X \to Y$ be a candidate limit function.

Remark (Indexed Families)**.** One can also consider a family $\{f_\alpha : X \to Y\}_{\alpha \in A}$. However, an arbitrary index set does not by itself define convergence. For example, equipping A with a directed order allows the family to be treated as a net. Here we focus on sequences.

Definition (Pointwise Convergence)**.** The sequence (f_n) converges pointwise to f on X if

$$f_n(x) \to f(x) \qquad \text{for every } x \in X.$$

Equivalently,

$$\forall x \in X \; \forall \varepsilon > 0 \; \exists N = N(x, \varepsilon) \in \mathbb{N} \; \forall n \geq N : \quad \rho(f_n(x), f(x)) < \varepsilon.$$

The index N may depend on x.

Definition (Uniform Convergence)**.** The sequence (f_n) converges uniformly to f on X if

$$\forall \varepsilon > 0 \; \exists N = N(\varepsilon) \in \mathbb{N} \; \forall n \geq N \; \forall x \in X : \quad \rho(f_n(x), f(x)) < \varepsilon.$$

The same N must work for every $x \in X$.

Remark (Supremum Criterion)**.** For nonempty X, uniform convergence is equivalent to

$$\sup_{x \in X} \rho(f_n(x), f(x)) \longrightarrow 0.$$

The supremum can initially be infinite, so it is viewed as an extended nonnegative real number.

The eventual pointwise bound $\rho(f_n(x), f(x)) < \varepsilon$ for all x gives only

$$\sup_{x \in X} \rho(f_n(x), f(x)) \leq \varepsilon.$$

To obtain a strict supremum bound below ε, apply uniform convergence with $\varepsilon/2$.

Remark (Quantifier Order)**.** The difference between pointwise and uniform convergence is the order of the quantifiers:

$$\begin{aligned} \text{Pointwise:} \quad & \forall x \; \forall \varepsilon > 0 \; \exists N \; \forall n \geq N, \\ \text{Uniform:} \quad & \forall \varepsilon > 0 \; \exists N \; \forall x \; \forall n \geq N. \end{aligned}$$

Uniform convergence implies pointwise convergence, but the converse need not hold.

Example (Pointwise but Not Uniform Convergence)**.** Define $f_n : [0,1] \to \mathbb{R}$ by

$$f_n(x) = x^n.$$

Then f_n converges pointwise to

$$f(x) = \begin{cases} 0, & 0 \le x < 1, \\ 1, & x = 1. \end{cases}$$

However, for every n,

$$\sup_{x \in [0,1]} |f_n(x) - f(x)| = 1.$$

Thus convergence is not uniform. Each f_n is continuous, but f is discontinuous at 1.

Theorem 11 (Uniform Limit Theorem)**.** *Let X, Y be metric spaces. If each $f_n : X \to Y$ is continuous and f_n converges uniformly to $f : X \to Y$, then f is continuous.*

Proof. Fix $x_0 \in X$ and $\varepsilon > 0$. By uniform convergence, choose N such that

$$\rho(f_N(x), f(x)) < \varepsilon/3 \qquad \text{for every } x \in X.$$

By continuity of f_N at x_0, choose $\delta > 0$ such that

$$d(x, x_0) < \delta \quad \Longrightarrow \quad \rho(f_N(x), f_N(x_0)) < \varepsilon/3.$$

For any x with $d(x, x_0) < \delta$, the triangle inequality gives

$$\begin{aligned} \rho(f(x), f(x_0)) &\le \rho(f(x), f_N(x)) + \rho(f_N(x), f_N(x_0)) + \rho(f_N(x_0), f(x_0)) \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\ &= \varepsilon. \end{aligned}$$

Therefore f is continuous at x_0. Since x_0 was arbitrary, f is continuous on X. □

Definition (Homeomorphic Spaces)**.** Two topological spaces X and Y are homeomorphic if there exists a homeomorphism $f : X \to Y$.

Definition (Homeomorphism)**.** A homeomorphism is a bijective function $f : X \to Y$ between topological spaces such that both f and its inverse f^{-1} are continuous. That is,

(a) f is a bijection

(b) f is continuous on X

(c) f^{-1} is continuous on Y

Homeomorphisms preserve topological properties, meaning that if two spaces are homeomorphic, they are considered topologically equivalent. This implies that they have the same topological structure, even if they may differ in other aspects such as geometry or metric properties.